Simple C++ Chatbot for Answering Basic User Queries

Mini Project Report

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Automata Computability and Complexity Course



Background: Chatbot Evolution and Types

Introduction to Chatbots

Used in support, healthcare, education

Rule-Based vs. AI-Based

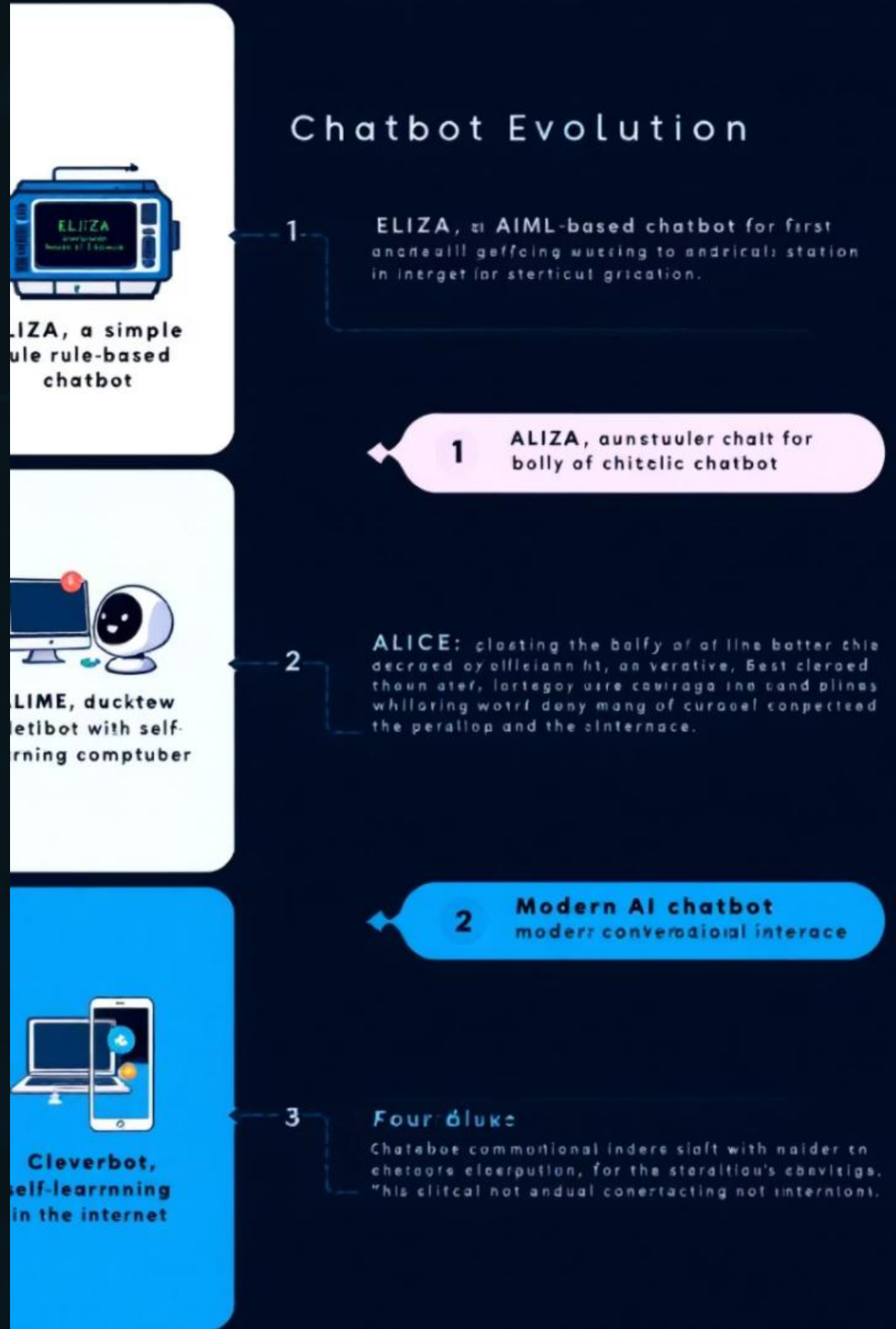
Rule-based use patterns; AI uses machine learning

Simple Educational Chatbots

Learn basic chatbot design easily

AI Adoption Statistic

75% customer interactions powered by AI by 2025



Project Overview

Project Aim

Build simple C++ chatbot for basic queries

Purpose

Understand string handling and conditionals in C++

How It Works

User input → response matching → output response

Made with GAMMA

Project Objectives

> This project aims to build a basic chatbot using C++ to handle simple queries, enhance programming skills in strings and conditionals, provide a user-friendly interface, and allow users to terminate the session with an "exit" command.

Create Basic Chatbot

Handle simple user queries effectively

Enhance Programming Skills

Master strings, conditionals, loops

User Interaction

Provide basic question and response interface

Exit Command

Type "exit" to terminate chat session





Tools & Technologies

Programming Language

C++ for performance and portability

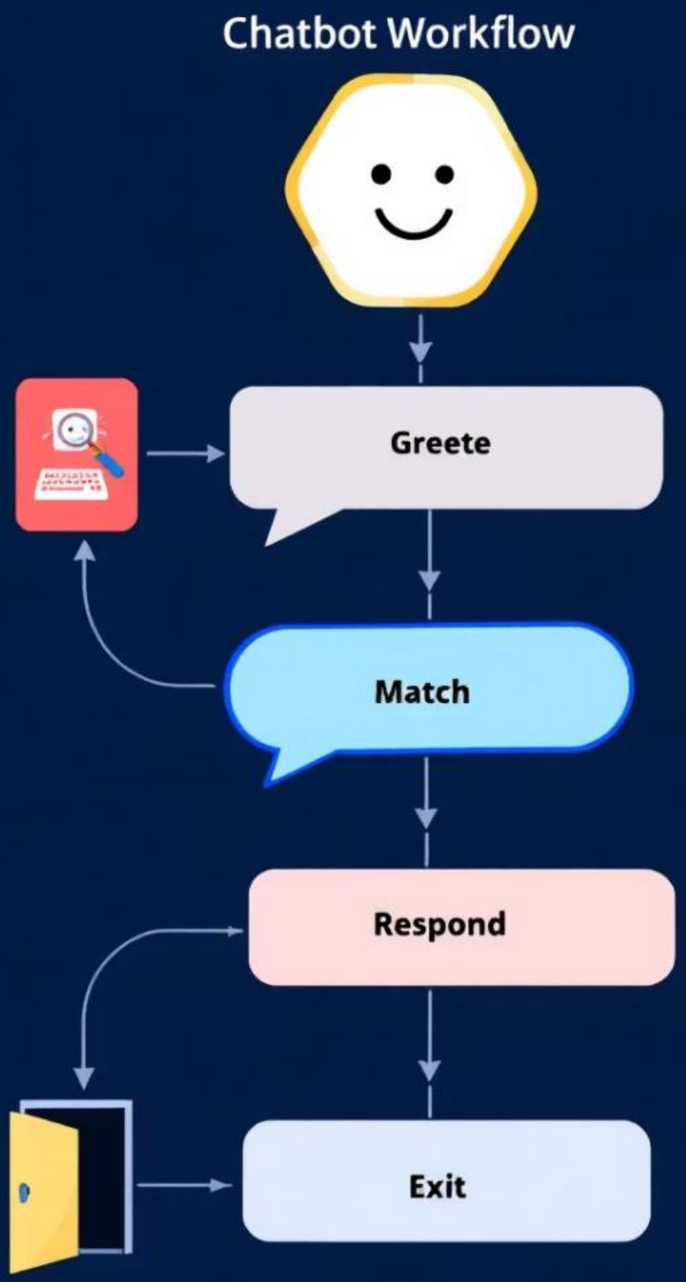
Key Libraries

- for input/output
- for string manipulation
- for utility functions

Development Environment

Code::Blocks or Visual Studio IDEs

Made with GAMMA



Project Workflow

Greet User

Initialize friendly welcome message

Await User Input

Listen for queries or exit command

Match Input

Compare with predefined responses

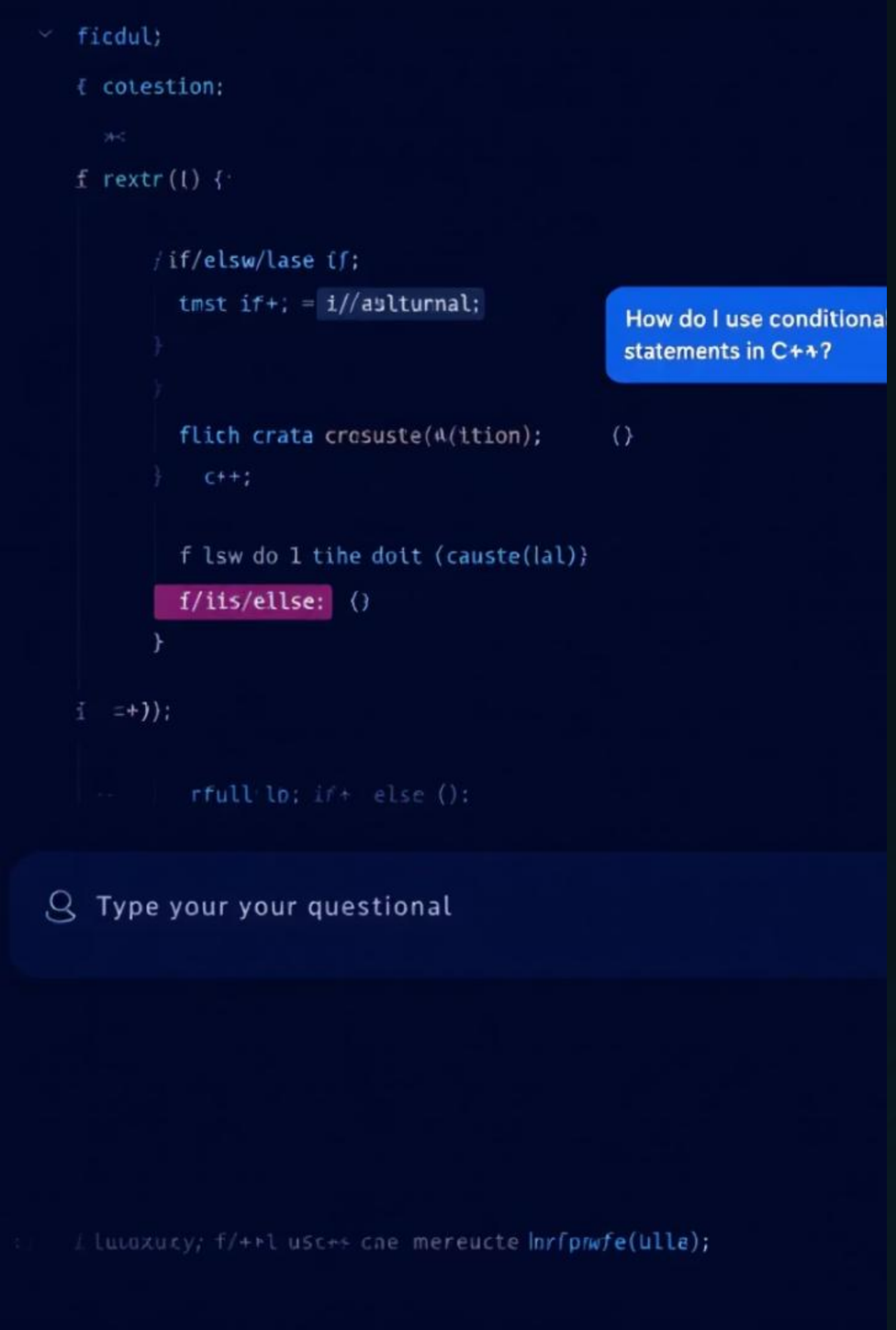
Exit Condition

Terminate chat when user types "exit"

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Code Snippet Example

```
if (userInput == "hello") {
    cout << "Hello! How can I assist you today?" << endl;
} else if (userInput == "what is your name?") {
    cout << "I am C++Bot, at your service!" << endl;
} else if (userInput == "exit") {
    cout << "Goodbye! See you next time." << endl;
    break;
}
```



Made with GAMMA

Results

Working Chatbot

Accurate responses for basic inputs

Skills Developed

Proficiency in strings, conditionals, loops

Deliverables

Fully functional chatbot program

Made with GAMMA



Future Enhancements

Advanced Interactions

Add trivia, jokes

Persistent Memory

Store previous interactions

GUI Integration

Use Qt or SFML for interface

NLP Integration

Understand complex queries

Made with GAMMA



Conclusion and Takeaways

Summary

Simple C++ chatbot demonstrates key programming concepts

Final Thoughts

Foundation for advanced chatbot systems

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